

Program : Diploma in Textile Technology	
Course Code : 4063	Course Title: Textile Colouration and Finishing
Semester : 4	Credits: 4
Course Category: Program core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- Impart thorough knowledge on colouring of Animal & Synthetic fibres and its blends.
- Outline the machineries used for Dyeing.
- To familiarize the techniques and methods of printing & finishing of textiles.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Textile wet Processing		Textile wet Processing	1
Textile Wet processing Lab I		Textile Wet processing Lab	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Discuss the preparatory process and colouring implementation for Animal & Synthetic fibres	14	Applying
CO2	Execute the dyeing of blended materials, Sketch the working of the dyeing machines suitable for dyeing textiles.	15	Applying
CO3	Implement different methods/styles to produce printed designs on the fabric.	14	Applying
CO4	Use of different finishing process and chemicals on textile materials to make it fit for particular end-use and effluent treatment	15	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2		3					3
CO3			3				
CO4				3	3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Discuss the preparatory process and colouring implementation for Animal & Synthetic fibres.		
M1.01	Application of dyes on Wool & Silk fibres	7	Applying
M1.02	Pretreatment and application of various dyes on Polyester fibres	7	Applying
Contents: Silk & Wool Dyeing: Preparation of wool and silk for dyeing - Various dyes used for protein fibres. Dyeing of wool with - Acid dyes - Basic dyes - Reactive Dyes - Dyeing of silk with - Acid dye - Basic dyes - Reactive dyes Polyester Pretreatments & Dyeing: Preparation of manmade material for dyeing - Dyeing of Polyester with disperse dyes - carrier, HTHP, Thermosol process.			
CO2	Execute the dyeing of blended materials, Sketch the working of the dyeing machines suitable for dyeing textiles.		
M2.01	Application of dyes on Polyester and cellulosic blended materials	5	Applying
M2.02	Application of various dyes on Polyester and wool blended materials	5	Applying
M2.03	Use of various Dyeing machines	5	Applying
	Series Test - I	1	
Contents: Blends Pretreatments & Dyeing: Explain the preparation of blended material for dyeing - Principles of Dyeing & Method of Dyeing blends - One bath and two bath dyeing process. Industry method of dyeing blends and checking washing fastness. Dyeing of P/C blends with - Disperse and vat dyes Dyeing of P/V blends with - Disperse and reactive dyes. Dyeing of P/W blends with - Disperse and acid dyes			

Dyeing Machinery: Classification - Principles - Loose cotton dyeing machine - Yarn package dyeing machine - Fabric Dyeing machines - Jigger - Winch - Padding mangles - Concept of Soft flow and Jet dyeing machine.

CO3	Implement different methods/styles to produce printed designs on the fabric.		
M3.01	Express the concept and significance of printing technology	3	Understanding,
M3.02	Outline the functions of various printing ingredients.	3	Applying
M3.03	Distinguish styles of printing	4	Understand
M3.04	Execute different methods of printing	4	Applying

Contents:

Printing: Definition and objectives - Differentiate dyeing & printing - Ingredients used in printing paste preparation - After treatments given to printed fabrics.

Styles of printing: Direct, Discharge, and Resist style

Methods of Printing: Study of different methods of printing - Block printing - Stencil Printing - Screen Printing - Automatic flatbed screen printing - Rotary screen printing - Roller printing - Transfer printing - Digital (ink-jet) printing - Sublimation Printing.

Conventional Methods - Batik Printing- Tie and dye - Foam printing - Flock printing - Damask printing - Kalamkari printing.

CO4	Use of different finishing process and chemicals on textile materials to make it fit for particular end-use and effluent treatment.		
M4.01	Summarize the process of finishing	4	Understanding
M4.02	Use of mechanical finishes to textile materials	4	Applying
M4.03	Application of chemical finishes given to textile materials	3	Applying
M4.04	Implement effluent treatment process for a dyeing plant.	4	Applying
	Series Test - II	1	

Contents:

Finishing: Objectives - Classification - Mechanical and Chemical finishes - temporary, semi-permanent and permanent finishing. Hydro extraction - Objectives of stentering - Working of Hot air stenter

Mechanical Finishes: Method of application - Calendaring – Objectives - Different types of calendars - Beetling - Sanforizing - Principle and mechanism of pre-shrinkage process - Description of sanforizing machine

Chemical Finishes: Anticrease finish - Study of Water repellent & water proof finishes - Method of testing the output - Study of fire retardant finishes - Method of testing the output. Defects in finishing & remedies.

Effluents treatments - BOD, COD - primary and secondary treatments process.

Text /Reference:

T/R	BookTitle/Author
T1	V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan publishers Ahmedabad.
R2	E.R Trotman - Dyeing & Chemical Technology of textiles
R3	R.S.Prayag - Dyeing of polyester and polyester blends
R4	R.R.Chakravarthy - A glimpses on the Chemical Technology of Textile Fibres
R5	J.T.Marsh - Textile Finishing
R6	D.G.Kale - Principles of Textile Printing

Online resources:

Sl.No	Website Link
1	http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers-according/ (accessed on 20 October 2013)
2	https://nptel.ac.in/courses/116/102/116102026/
3	https://nptel.ac.in/courses/116/102/116102016/
4	https://swayam.gov.in/
5	http://eacharya.inflibnet.ac.in/vidya-mitra/